On a scale accessible to studies of modern organisms, drainages of water, even of small streams, are divisions, but over time, with regions uplifting and subsiding, the millions of small divisions change. Climate changes, soils change, rock subject to erosion changes, volcanic ash buries forests—all sorts of things happen. This is the milieu of constant pressure on all living organisms that forces adaptive change and creates new species. At its core, all this change, including that of the organisms, is geological in origin. Darwin, accepting Lyell's interpretation of geology, saw the fundamental contradiction that underlay Lyell's interpretation of the fossil record. In the end, he was truer to Lyell's vision than was Lyell himself. The Galápagos Islands are a microcosm of island-by-island divisions between ecosystems. They caught Darwin's attention, they underscored the difficulties with Lyell's hypothesis of organic change, and they helped Darwin to formulate a new theory. The Galápagos Islands guide evolutionary studies to this day.

SEE ALSO THE FOLLOWING ARTICLES

Cape Verde Islands / Coral / Galápagos Islands, Geology / Plate Tectonics / Volcanic Islands / Wallace, Alfred Russel

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DARWIN'S FINCHES

SEE GALAPAGOS FINCHES

DECEPTION

SEE ANTARCTIC ISLANDS

DEFORESTATION

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Deforestation is the loss of trees involving a vegetational succession from forest cover to some other kind of landscape. It causes reduced biodiversity and, under certain conditions, promotes the development of wastelands. The process of deforestation is intertwined with human history as well as with the influence of climate and other environmental factors. Islands are particularly vulnerable to deforestation because their ecosystems tend to be fragile and susceptible to rapid change.

PAST AND PRESENT EVIDENCE FOR DEFORESTATION

Historical records offer abundant evidence for deforestation, especially in relation to the loss of trees through the development of agriculture and timber industries. Records for the Philippines show, for example, that primary-growth forest cover decreased from about 70% to 7% of the total land area over the past 100 years. For islands without such records, or where deforestation occurred in the distant past, there are a number of lines of evidence that can help to identify the occurrence and extent of deforestation. Pollen profiles reconstructed from sediment cores are particularly revealing. Typical pollen profiles indicating deforestation show declining frequencies of tree pollen with simultaneous increases in charcoal and the pollen of grasses. The frequency and size of charcoal particles in the sediments may indicate the role of fire in the loss of tree cover.

Large areas of habitually burned grasses or ferns are another common indicator of deforested landscapes (Fig. 1). Grass- or fern-dominated vegetation forms after burning and is maintained by fire that inhibits the growth of woody competitors. This is the case in New Zealand, where most of the lowlands became deforested following the arrival of Polynesian settlers, and forests were replaced by vast areas dominated by ferns.

Geomorphology can also serve as a marker for deforestation, especially on islands with steep terrain. By removing the tree canopy, deforestation exposes ground surfaces to the direct impact of rainfall. This promotes erosion, which can be accentuated by various factors including slope gradient, rainfall levels, severe weather, and the type of groundcover. In extreme cases, massive erosion results in completely denuded landscapes scarred by deep gullies.

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FIGURE 1 View of Hanamiai Valley, Tahuata, Marquesas Islands (French Polynesia). *Miscanthus* grasslands, a form of vegetation maintained by fire, are visible as light-colored areas on deforested ridges leading to the central mountain range. The valley floor is an example of forest replacement, where native trees have been replaced by economically important introduced species of trees. Photograph by B. Rolett, July 1998.

Sediments eroded from slopes accumulate at the base of these slopes and may also be transported by water to valley floors, coastal lowlands, and inshore marine environments. Many islands display physical evidence for erosion and sedimentation, and in certain tropical settings deposits up to 5 m deep have accumulated in less than 1000 years.

ENVIRONMENTAL PREDICTORS OF ISLAND DEFORESTATION

Comparative studies of islands help to identify environmental factors contributing to deforestation. Small, dry, cool, remote islands, with low topography, low inputs of soil-enriching continental dust, and little or no volcanic ash fallout are the most vulnerable to deforestation. Environmental factors aiding the retention of forest cover include high rainfall, large island size, and high levels of volcanic ash. Dry islands are predisposed to deforestation because rainfall is a primary determinant of plant growth rate. Low rainfall also increases forest vulnerability to fire. By contrast, high rainfall favors plant growth and decreases the chance of deforestation.

Large, high islands generally retain more forest than do small, low islands. This is particularly true on islands with tracts of mountainous land that are inaccessible to humans and their fires. Large islands may also have greater habitat and tree species diversity, increasing the likelihood that some species will survive. Furthermore, large islands create their own rain (orographic rainfall) when moist air blowing from the ocean hits mountains and rises until it condenses. Geological age is relevant because soil nutrients become lost from older formations with time, especially by rain leaching. Lost nutrients can be restored by volcanic ash and continental dust fallout. Islands receiving little or no aerial dust or ash are thus more vulnerable to deforestation than are islands closer to volcanoes ejecting tephra and ash. Guano from nesting birds enriches soils on islands where birds are or were common, but it is difficult to quantify the importance of this factor.

THE HUMAN ELEMENT

Although geography is important, cultural practices also have guided the contrasting ecological histories of different islands. Human populations play a direct role in deforestation by clearing land for gardens, cutting trees to obtain timber and firewood, and so on. The importance of humans in these activities varies significantly over time and from one island to another depending upon factors including population size and the scale of the economy. On many forested islands, native species of trees have been replaced by economically valuable introduced species. This process of forest replacement can transform the species composition of the forest without greatly altering the ecology. A classic example is Tahiti, where Polynesian settlers introduced many different species of fruit trees to thoroughly transform the character, but not the ecology, of the island's lowland forests. Captain Cook and other early European explorers mistook these bountiful anthropogenic forests for unaltered environments, and this view contributed to the European idealized image of Tahitians as a people living in harmony with nature.

FIRE

Because most fires are set by people, human colonization introduces fire as a powerful force in island deforestation. It is particularly significant in the deforestation of dry islands. Traditional agricultural systems often rely on fire for clearing fields, and on dry islands there is a high risk of fires spreading from gardens and burning out of control. When this occurs repeatedly, forests are replaced by savannahs of grasses or ferns, which can survive repeated firing. On dry islands throughout the eastern Pacific, fire-associated grasslands unsuitable for cultivation cover large tracts of land.

THE ROLE OF INTRODUCED ANIMALS IN DEFORESTING ISLANDS

Most islands have impoverished native floras and faunas as a consequence of their isolation. When this isolation is broken by the human introduction of alien species, the environmental consequences are highly disruptive. Introduced grazing animals, especially goats and sheep, are particularly destructive. They inhibit forest regeneration by grazing on young saplings, while also exacerbating

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soil erosion through trampling the ground with their hard hooves. Goats and sheep were introduced (together with agriculture) to the Mediterranean during the Neolithic era, and they are clearly implicated in the deforestation of this region. Yet because people also played a direct role in deforesting the Mediterranean islands, it is difficult to gauge the relative importance of grazing animals, as opposed to humans, in this process.

There are a number of examples in which sheep and goats were left alone on uninhabited islands, and in all cases the outcome was an environmental disaster. Two of these islands are in the Marquesas, a South Pacific archipelago of ten high volcanic islands known for rugged, verdant landscapes. When these islands were first discovered, by Polynesians around 1200 years ago, terrestrial mammals were absent and had never existed in the Marquesas. Seafaring Polynesians introduced pigs, dogs, rats, and chickens, in addition to numerous plants, including root crops, fruit trees, and other cultigens. Human colonization and the introduction of alien species led to the mass extinction of native landbirds, an event that surely reflects broad environmental shifts during the first centuries of Polynesian settlement. Nevertheless, Marquesans developed a flourishing culture, and they achieved sustainable growth for more than 1000 years, until the dawn of European contact.

When Europeans and Americans began to frequent the Marquesas during the late eighteenth century, they initiated a new set of experiments involving the introduction of terrestrial animals. Ship captains released breeding pairs of sheep and goats in various locations across the Marquesas. Goats quickly became feralized throughout the archipelago, whereas herds of sheep sprang up on the uninhabited islands of Eiao and Mohotane. In contrast to the large valleys, deep bays, and accessible coasts of the principal islands, Eiao and Mohotane are dominated by dry plateaus, and their coastlines consist mainly of cliffs. The seamen who introduced these animals could hardly have anticipated the long-term environmental consequences. Both islands supported dryland forests and perennial streams when sheep and goats were introduced. Marquesan forests typically have a canopy of tall trees rising above a dense understory. Today on Mohotane, the stands of tall trees are dwindling, and grazing has completely eliminated the understory (Fig. 2). As the surviving trees reach the end of their life cycle, the forest is vanishing, and as this occurs, the ground is exposed to the direct impact of rainfall. Heavy rains and runoff have produced deep gullies cutting down to lateritic clays and lava bedrock.

The situation is similar but worse on Eiao, another island in the Marquesas (Figs. 3, 4). Eiao's plateau is a red



FIGURE 2 Stands of surviving trees on Mohotane, Marquesas Islands (French Polynesia). Grazing by sheep, introduced less than 100 years ago, has eliminated the understory and has largely deforested the island. Photograph by B. Rolett, April 1987.



FIGURE 3 View of Eiao, Marquesas Islands (French Polynesia), deforested by sheep and goats introduced less than 200 years ago. One of the few surviving trees (at right) is near the end of its life cycle. Photograph by B. Rolett, December 1999.



FIGURE 4 Close-up of gullies produced by erosion and runoff on the plateau of Eiao, Marquesas Islands (French Polynesia), as a result of deforestation. Photograph by B. Rolett, December 1999.

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clay desert, and the only surviving trees live in sheltered valleys that trap soils washing off the surrounding slopes. The dense understory in these places denies access to the sheep, preserving small patches of vegetation on an otherwise denuded island. However, Eiao can be contrasted with its close neighbor Hatutaʻa, which has an equally fragile landscape. Not only is Hatutaʻa fully forested, but it also is considered one of the least disturbed tropical dry forests in Polynesia. Although it is located only a few miles from Eiao, Hatutaʻa has largely escaped the attention of humans. Polynesians visited but never settled the island, which has no bays or good landings. The Pacific rat is the only introduced animal from any period in history, and the forests have survived nearly unaltered. All evidence suggests that Eiao's original vegetation, now entirely lost, was similar to the forests still present on neighboring Hatutaʻa. The harshly contrasting environmental histories of Eiao and Hatutaʻa are mainly attributable to the introduction of sheep and goats. This comparison illustrates the inherent fragility of island environments, while also highlighting the dangerous role of grazing animals in deforestation.

AVOIDING DEFORESTATION

Although islands are susceptible to rapid environmental change after the breaching of their natural isolation, different islands have experienced varying rates and degrees of deforestation. The different outcomes can be explained by a combination of environmental and cultural factors. Forest retention is usually greater on islands where people developed alternatives to fire-intensive forms of agriculture. The most common fire-intensive form of agriculture is shifting cultivation, in which forested land is cleared by burning. In shifting cultivation, gardens are planted for a series of cropping cycles, and then the fields are allowed to lie fallow before they are cleared and burned again to prepare them for a new cropping cycle. One cultural response that favors the retention of forests is the development of irrigated agriculture as an alternative to shifting cultivation. Irrigated fields (with crops such as rice or taro) generally produce higher yields than rainfed gardens do, so irrigated agriculture conserves forests by limiting the amount of arable land needed for food production. In addition, irrigated agriculture does not require the use of fire. This is an advantage on dry islands, where fires can burn out of control. The development of arboriculture also favors the retention of forests. People who rely on arboriculture are not forced to cut trees in order to expand their subsistence economy, because the forest itself is a primary source of food. Instead, in arboriculture people transform forests without destroying them, by replacing native trees with useful species such as breadfruit, chestnuts, and olives. Other cultural responses by people living on islands also favor the retention of forests. These include controlling human population growth, conserving resources, and limiting the introduction of non-native species, especially grazing animals.

SEE ALSO THE FOLLOWING ARTICLES

Climate on Islands / Human Impacts, Pre-European / Lava and Ash / Pigs and Goats / Sustainability / Vegetation

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DISEASE

SEE BIRD DISEASE, PLANT DISEASE

DISPERSAL

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Dispersal (or migration) can be defined as any movement of individuals with potential consequences for gene flow across space. Dispersal can be extended to include movement through time (i.e., dormancy). Dispersal is a fundamental life history trait, with multiple demographic and genetic consequences. In particular, it determines the amount of local adaptation and the likelihood of various speciation mechanisms. It is also an evolving trait.

EVOLUTIONARY CAUSES OF DISPERSAL

Overall, the different causes for dispersal evolution can be classified into three broad classes of explanation. First, the evolution of dispersal may be understood in the light of habitat selection theory, whereby behaviors at departure or arrival help individuals to preferentially establish in certain (supposedly better) types of habitat. In particular, by allowing escape from crowding or inbreeding, dispersal, whether or not triggered by density or local relatedness, may help individuals to reach

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